

Business Problem Statement

A national retail chain sells office supplies, furniture, and technology products across four U.S. regions. Leadership has noticed that overall revenue has grown steadily year over year, but profit growth has not kept pace — and in some product lines and states, the company is losing money outright despite strong sales volume.

Management does not have a clear, data-backed view of where profit is actually being made or lost, which customers are most valuable, or which discounting practices are helping versus hurting the bottom line. Decisions about discount policy, regional strategy, and customer retention are currently being made without this visibility.

Overarching Business Question

“Where is the company’s profit actually coming from, where is it being lost, and how can data guide smarter decisions on discounting, regional strategy, and customer retention?”

Deliverables

1. Data Preparation & Modeling (Python): Clean and transform four years of raw order-level transaction data (2014–2017); engineer features needed for time-based and profitability analysis.
2. Data Analysis (SQL): Load the cleaned dataset into PostgreSQL; write structured queries covering product/customer ranking, time trends, cohort retention, and RFM customer segmentation.
3. Visualization & Insights (Power BI): Build an interactive dashboard connected live to the PostgreSQL database, allowing stakeholders to filter by region, category, and segment.
4. Report and Presentation: Summarize key findings and business recommendations in a clear written report.
5. GitHub Repository: Publish the Python notebook, SQL scripts, and dashboard file in a well-structured public repository as a portfolio piece.

Key Questions to Answer

- Which product sub-categories and states are profitable, and which are structurally losing money?
- Does discounting help or hurt profit, and at what threshold does it start causing losses?
- How does demand change across months and seasons, and is that seasonality consistent year over year?
- Which customers are most valuable, and which valuable customers are at risk of churning?
- How does profitability vary by region and by customer segment (Consumer, Corporate, Home Office)?